LeRobot Installation Manual

Installing LeRobot on another Mac

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This manual explains how to install LeRobot, the OCaml drawing robot, on a Mac where it has never been installed. Once it is done, use the *LeRobot User Manual* (`LeRobot_Manual.pdf`) to draw and to run robot programs.

The installation takes about half an hour, mostly spent waiting for downloads. You need:

- a Mac with a recent macOS (Apple Silicon or Intel);
- an administrator account: several steps ask for the Mac password;
- an Internet connection;
- a copy of the `LeRobot` folder (USB key, AirDrop, cloud drive, ...).

All commands are typed in the **Terminal** application (*Applications* → *Utilities* → *Terminal*). Copy each command in one piece, paste it into the Terminal, and press Return. When `sudo` asks for a password, type the Mac password: nothing is shown while you type, which is normal.

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1 Apple's command line tools

These provide the C compiler and `make` that OCaml needs.

```
xcode-select --install
```

Click *Install* in the window that appears and wait for it to finish. If the Terminal answers that the tools are already installed, go on to the next step.

2 Homebrew

Homebrew installs the other programs. Skip this step if `brew -version` already prints a version number. Otherwise run the installer (the same command is given on `https://brew.sh`, where it can also be copied):

```
/bin/bash -c \
  "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

At the end, the installer prints a few *Next steps* commands. Run them: they make the `brew` command available in every new Terminal window. On an Apple Silicon Mac they amount to:

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
```

3 XQuartz

XQuartz is the display server in which the robot window opens.

```
brew install --cask xquartz
```

Then **log out of the Mac and log in again** (Apple menu → *Log Out*). XQuartz only works properly after a new session.

4 OCaml, dune and the graphics library

1. Install opam, the OCaml package manager:

   ```
   brew install opam
   ```

2. Initialise it. This downloads and compiles OCaml, which takes several minutes. Answer `y` if it asks whether to modify the shell configuration.

   ```
   opam init -y && eval $(opam env)
   ```

3. Install the build system and the libraries LeRobot uses:

   ```
   opam install -y dune graphics ocamlfind && eval $(opam env)
   ```

Check: `ocaml -version` and `dune -version` each print a version number.

5 The LeRobot folder

1. Copy the `LeRobot` folder anywhere, for example onto the **Desktop**. The OCaml programs are in its folder `LeRobot_OCaml_FR`; the `LeRobotOCaml.app` icon must stay in that folder, next to `robot.sh`. The whole `LeRobot` folder can be moved or renamed later: nothing in it depends on where it is.

   The `LeRobot` folder can be anywhere (on the Desktop, in Documents, ...): everything in it is found relative to it. The commands of this manual are typed in a Terminal window *opened in the `LeRobot` folder*: type `cd` and a space, drag the `LeRobot` folder from the Finder onto the Terminal window, and press Return (or, in the Finder, right-click the folder and choose *Services* → *New Terminal at Folder*).

2. Compile it:

```
cd LeRobot_OCaml_FR && eval $(opam env) && dune build
```

If the folder was copied with its `_build` subfolder from another Mac and the build fails, delete it with `dune clean` and build again.

6 Start XQuartz automatically after each restart

At every restart, macOS empties the folder `/tmp`, which removes the folder `/tmp/.X11-unix` that XQuartz needs. XQuartz cannot recreate it itself, because it must belong to the system administrator (root). Without it, XQuartz opens and closes at once and the robot window cannot appear.

The file `org.xquartz.x11-unix.plist`, in the `LeRobot` folder, is a small startup task that recreates this folder at every restart. Install it once with these four commands. Copy the four lines together and paste them into the Terminal; the first one asks for the Mac password.

```
sudo cp org.xquartz.x11-unix.plist /Library/LaunchDaemons/
sudo chown root:wheel /Library/LaunchDaemons/org.xquartz.x11-unix.plist
sudo chmod 644 /Library/LaunchDaemons/org.xquartz.x11-unix.plist
sudo launchctl bootstrap system /Library/LaunchDaemons/org.xquartz.x11-unix.plist
```

The last command also runs the task at once, so no restart is needed. If the `.plist` file is missing from the folder, recreate it from Appendix A first.

Check that it worked:

```
ls -ld /tmp/.X11-unix
```

The answer must begin with `drwxrwxrwt` and contain `root wheel`.

If you ever want to remove the startup task (do not run this to install it):

```
sudo launchctl bootout system/org.xquartz.x11-unix
sudo rm /Library/LaunchDaemons/org.xquartz.x11-unix.plist
```

7 First launch

1. Start XQuartz: `open -a XQuartz`.
2. Open the interactive drawing program:

   ```
   cd LeRobot_OCaml_FR
   eval $(opam env)
   DISPLAY=:0 dune exec ./Dessiner.exe
   ```

   The robot window (800 × 700 pixels) opens in XQuartz. If you do not see it, click the XQuartz icon in the Dock. Pressing Return or clicking in the window ends the program.
3. `DISPLAY=:0` must always be put in front of the command: the default display setting may point to a stale XQuartz connection. The shortcut `make Dessiner`, typed in the same folder, does all of this.
4. `LeRobotOCaml.app` starts XQuartz if it is not running. Double-click its icon to run `Dessiner`; drop a `.ml` program on its icon to compile and run it. The first time you open `LeRobotOCaml.app`, macOS may refuse to open it because it comes from another Mac. Right-click the icon, choose *Open*, then confirm. Allow it to control Terminal and to access the folder of `LeRobot` (Desktop, Documents, ...) when macOS asks.

8 Troubleshooting

Symptom	Cause	Fix
command not found: brew	Homebrew's *Next steps* were not run	Run the `echo 'eval ...'` command of Section 2, or open a new Terminal window
command not found: dune or `ocaml`	opam's settings not loaded in this window	Run `eval $(opam env)`
`opam install graphics` fails, mentioning X11	XQuartz not installed, or the Mac not logged out and in again	Do Section 3 completely, then retry
XQuartz opens and closes at once	`/tmp/.X11-unix` missing or not owned by root	Run the startup task again: `sudo launchctl kickstart system/org.xquartz.x11-unix`; if that fails, install it (Section 6)
Cannot open display	`DISPLAY=:0` missing, or XQuartz not running	Put `DISPLAY=:0` in front of the command; `open -a XQuartz`
Boot-out failed: 3: No such process	The removal command was run while the task was not installed	Harmless; run the installation commands of Section 6
Bootstrap failed: 5	The task is already installed	Nothing to do
XQuartz log	Detailed error messages	`~/Library/Logs/X11/org.xquartz.log`

A The startup task file

If `org.xquartz.x11-unix.plist` is missing, this command recreates it in the `LeRobot` folder (the Terminal being opened in that folder); then install it as in Section 6.

```
cat > org.xquartz.x11-unix.plist <<'EOF'
<?xml version="1.0" encoding="UTF-8"?>
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>org.xquartz.x11-unix</string>
  <key>ProgramArguments</key>
  <array>
    <string>/bin/sh</string>
    <string>-c</string>
    <string>mkdir -p /tmp/.X11-unix
chown root:wheel /tmp/.X11-unix
chmod 1777 /tmp/.X11-unix</string>
  </array>
  <key>RunAtLoad</key>
  <true/>
</dict>
</plist>
EOF
```